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Information Presentation Response

After learning about Weapons of Math Destruction, my perspective on data-driven decision-making has changed, particularly regarding how algorithms reinforce societal inequalities. Previously, I believed that data-driven systems were inherently objective, but the book highlights how algorithms embed biases, such as in predictive policing and hiring discrimination (O'Neil, 2016). Additionally, ProPublica's research shows that risk assessment algorithms in the criminal justice system disproportionately classify Black defendants as "high risk" (Angwin et al., 2016). If future developments ensure greater transparency in algorithm design and stronger regulatory oversight, my views on algorithmic fairness may evolve.

Algorithmic University Admissions

The answer for this question is that we should exclud **biased and ethically problematic data** from POST. Including certain data points in the Program to Optimize Student Tuition (POST) algorithm could introduce significant bias, leading to unfair and potentially discriminatory admissions decisions. While machine learning can help identify successful candidates, it's important to exclude information that could exacerbate inequalities.

The first thing is that we should exclude **race and gender** to prevent discriminatory bias. Including race and gender could lead to biased admissions decisions, as seen in Amazon's hiring algorithm, which penalized women due to historical biases (Dastin, 2018). Machine learning models trained on past inequalities tend to perpetuate discrimination rather than eliminate it (O'Neil, 2016). To maintain fairness, these factors should not be used in the admissions model.

The second thing is that we should reduce the proportion of considerations of **family income and parents' educational history**. Using family income or parental education history as predictors can disproportionately benefit students from wealthier backgrounds while disadvantaging those from lower-income families. Studies have demonstrated that algorithms incorporating socioeconomic data can amplify disparities rather than leveling the playing field (Clark, 2023). In criminal justice, risk assessment algorithms have unfairly classified individuals from low-income backgrounds as high risk, showing how socioeconomic bias can be embedded in predictive models (Angwin et al., 2016). Applying similar principles to university admissions could lead to an unfair advantage for students from privileged backgrounds.

Thirdly, we should exclude **criminal history** from the model to avoid reinforcing historical inequities. Including criminal history in the admissions model could disproportionately disadvantage certain racial and socioeconomic groups due to documented biases in the criminal justice system. ProPublica's analysis of risk assessment tools have shown that such systems frequently misclassify Black defendants as high risk while underestimating the risk for white defendants (Angwin et al., 2016). Using this type of data for university admissions would unfairly penalize applicants based on external systemic biases rather than academic potential.

Apart from above, we also should exclude **address and taxation history**. Geographic and tax history often serve as proxies for race and socioeconomic status. Mortgage lending algorithms have been shown to discriminate against minority neighborhoods due to these correlations (Clark, 2023). Using address data in admissions could lead to similar bias, indirectly favoring applicants from wealthier areas.

Consequently, to build an ethical admissions system, race, gender, socioeconomic background, criminal history, and geographic data should be excluded. These factors introduce bias, undermining the goal of selecting students based on ability. Instead, the model should prioritize academic and extracurricular achievements. Without proper safeguards, the POST system risks becoming a “weapon of math destruction” rather than a tool for fairness (O’Neil, 2016).

Common Crawl

While Common Crawl’s large-scale web scraping provides diverse training material, its data selection and filtering introduce biases that affect AI models. These biases can reinforce dominant narratives, marginalize underrepresented voices, and spread misinformation.

Firstly, we consider the bias in **data selecting and filtering**. Common Crawl prioritizes high-traffic and frequently updated websites, leading to an overrepresentation of mainstream sources and an underrepresentation of niche perspectives (Dodge et al., 2021). Major news outlets, Wikipedia, and corporate websites dominate, reinforcing Western-centric narratives (Bender et al., 2021). Additionally, since Common Crawl does not distinguish between reliable and unreliable sources, misinformation can persist, affecting AI model accuracy (Brown et al., 2020).

Also, there exist some biases in **language and culture**. A significant limitation of Common Crawl is its heavy bias toward English-language content, with disproportionately low representation of non-English sources (Brown et al., 2020). Since generative AI models rely on frequency and pattern recognition, an overrepresentation of one language leads to better fluency and accuracy in English while diminishing the model’s ability to handle fewer common languages. This creates a disparity in AI accessibility, where users from non-English-speaking regions receive lower-quality responses, grammatical inaccuracies, or culturally insensitive content (Bender et al., 2021). Additionally, the dominance of Western sources skews AI-generated content toward Eurocentric viewpoints, limiting cross-cultural representation.

In the meanwhile, the common crawl might also p**erpetuate the harmful stereotypes**. Common Crawl includes unfiltered social media posts, forum discussions, and user-generated content, which frequently contain gender, racial, and socioeconomic biases. AI models trained on such data have been found to replicate and even amplify sexist, racist, and discriminatory language patterns (Dodge et al., 2021). For example, studies show that biased training data leads to discriminatory hiring AI tools that unfairly disadvantage female applicants (Dastin, 2018). Similarly, biased law enforcement data in machine learning models has been linked to racial profiling and unequal treatment in predictive policing (Angwin et al., 2016). Without careful filtering and bias mitigation strategies, AI risks reinforcing rather than challenging societal biases.

There are also some problems about the **content verification**. Since Common Crawl does not verify sources, unreliable blogs, misleading opinion pieces, and conspiracy theories can be scraped and included in AI training data (Brown et al., 2020). This makes AI-generated content susceptible to hallucination—producing inaccurate or misleading information. If AI models lack mechanisms to distinguish between fact-checked journalism and unverified claims, they may amplify misinformation, making them unreliable for critical applications like news summarization or academic research (Bender et al., 2021). This issue is particularly concerning in areas such as healthcare, law, and politics, where AI-generated misinformation can have severe consequences.

To **reduce the biases** inherent in Common Crawl, AI developers should implement improved data filtering techniques that prioritize reliability and inclusivity. This includes increasing human oversight in dataset curation, incorporating underrepresented languages and cultures, and applying bias detection algorithms to mitigate harmful patterns (Dodge et al., 2021). Additionally, transparency in dataset usage—documenting the sources and selection criteria—would help ensure AI models remain accountable and fair (Bender et al., 2021).

Consequently, Common Crawl plays a vital role in AI training, but its biases in data selection and filtering impact AI-generated content. Overrepresentation of English, mainstream media, and unreliable sources leads to misinformation, cultural bias, and stereotype reinforcement. Addressing these issues is essential for building more ethical and inclusive AI systems.

Reference

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